Shang-Jui Ray Kuo

Curriculum Vitæ

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Research Interests

My research focuses on fundamental questions about the architectural choices AI systems have inherited and built on. Current work targets the input interfaces of vision-language models, with concurrent threads on both sides: state space models (Mamba family) as vision encoders, and learned alternatives to tokenizer + embedding lookup for languages the standard pipeline was never designed for. My background in hardware and systems lets me work close to systems to recognize where the original constraints came from, while standing firmly on the AI side to propose alternatives the broader community can build on.

Education

Stony Brook University — Stony Brook, NY
Ph.D. in Computer Science — *Aug. 2024 – Spring 2029 (expected)*

- **Advisor:** Prof. Paola Cascante-Bonilla, SPELL Lab (Synthetic Perception and Learning Lab)
- **Selected coursework:** CSE 537 Artificial Intelligence (Spring 2025, Prof. Niranjan Balasubramanian), Topics in CS: Vision-Language Foundation Models (Fall 2025), CSE 538 Natural Language Processing (Spring 2026, Prof. Niranjan Balasubramanian)

National Taiwan University — Taipei, Taiwan
B.S. in Electrical Engineering — *Sept. 2019 – June 2023*

- **GPA:** 3.62/4.3 overall; 4.02/4.3 last 60 credits
- **Selected coursework:** Deep Learning for Computer Vision*, Machine Learning*, Digital Signal Processing in VLSI Design*, Digital System Design, Integrated Circuit Design, Computer Architecture, Algorithms, Data Structures (* = graduate-level)

Publications

Peer-reviewed conference publications

Improving Limited Supervised Foot Ulcer Segmentation Using Cross-Domain Augmentation Strategies

- **Shang-Jui Kuo***, Po-Han Huang*, Chia-Ching Lin, Jeng-Lin Li, Ming-Ching Chang. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024. IEEE Xplore *(* = equal contribution)*

Under review

Do VLMs Need Vision Transformers? Evaluating State Space Models as Vision Encoders

- **Shang-Jui Ray Kuo**, Paola Cascante-Bonilla. *Under Review*, 2026. arXiv | Project Page | Code | HF Paper | Checkpoints
- Featured on Hugging Face Daily Papers; HF open-source team offered a ZeroGPU (A100) grant for a public demo; poster accepted to SUNY AI Symposium 2026.

Research Experience

Graduate Researcher — SPELL Lab — Sept. 2025 – Present
Stony Brook University | Advisor: Prof. Paola Cascante-Bonilla — *Stony Brook, NY*

- Designed a strictly controlled LLaVA-style backbone-swap (frozen vision tower, fixed Vicuna-7B + 2-MLP connector, matched ImageNet-1K/224^2 initialization) and showed VMamba (pure SSM, ∼30–89M params) leads ViT-family backbones up to ∼662M params on RefCOCO / RefCOCO+ / RefCOCOg grounding benchmarks while remaining competitive on open-ended VQA
- Showed ImageNet accuracy and naive backbone scaling are unreliable predictors of downstream VLM performance — a result that should reshape how the community selects vision towers
- Diagnosed “localization collapse” in some detection-pretrained checkpoints as a vision–language interface failure (not architectural); proposed simple stabilizations (3-MLP connector + square input geometry) recovering collapsed localization to near-baseline

Research Assistant — COMPAS Lab — Sept. 2025 – Dec. 2025
Stony Brook University | Advisors: Prof. Mike Ferdman, Prof. Peter Milder — *Stony Brook, NY*

- Retargeted an existing PyTorch-to-NPU pipeline (custom RTL NPU executing the lab’s ML-ISA framework natively) to AMD AI Engine cores on the VCK5000 FPGA via runtime-only modifications — demonstrating the framework’s hardware-portability claim across heterogeneous accelerators

- Invented an RTL-level on-chip debug methodology with configurable debug registers under broken-toolchain conditions (no working hardware simulator; 8-hour bitstream compile cycles); first working matmul kernel on AIE in the lab, establishing the integration pattern for future ML-ISA-AIE work

Toward Tokenizer-Free Chinese Language Models CSE 538 NLP, Spring 2026 – ongoing

- Originated as CSE 538 NLP final project (solo technical execution); continuing as a PhD research direction. Built a controlled comparison of three Chinese character-embedding families (char-ID lookup, IDS radical decomposition, rendered-glyph vision encoders) under a fixed ModernBERT-small backbone; structure-aware variants match the char-ID baseline on C-MTEB (31 frozen tasks); introduced a low-cost frozen-encoder pixel-reconstruction probe for OOV-character behavior
- First concrete milestone of a longer PhD research program reframing tokenization as a *learned token-construction* problem — sister work to the VLM-SSM paper on the vision side of the same input-interface thread

Adaptive Chess Tutor: LLM Dual-LoRA Fine-tuning CSE 537 AI, Spring 2025

- Trained TinyLlama-1.1B as a locally-deployable chess tutor combining move prediction and human-like explanation in one model via dual-LoRA: programmatic SFT (60K Lichess-derived samples via `python-chess` verifier) lifted `can_piece_move` accuracy from 2.9% to 94.2% over base; cognitive distillation from a Mixtral-8x7B teacher (10K samples) into a second LoRA layered on top of a frozen Phase 1 adapter
- Runtime α/β weighted-adapter blending via PEFT's `add_weighted_adapter` gives a reproducible 17-point trade-off curve between move quality (Stockfish Score Delta) and explanation similarity (BERTScore F1), characterized across 47 evaluations on the NuWulf cluster. Stack: PEFT, QLoRA 4-bit (bitsandbytes), HF Transformers, BERTScore, SLURM on H200/B40 GPUs.

Industry Experience

AI Researcher & AI Accelerator Engineer Mar. 2023 – Feb. 2024
Inventec Corporation — AI Center & Digital Center (AI on Chip) *Taipei, Taiwan*

- Contributed to a commercial NPU IP product (VectorMesh™) during initial commercialization: designed RTL image-resize DSP module + built NPU co-simulation and verification environment. Product later won the EETimes 2024 Asia Golden Award for Best IP/Processor (after my departure)
- Designed a lightweight CNN (10–50K params) for stylus trajectory regression from capacitive sensors, achieving sub-2-pixel accuracy and ∼50% error reduction; deployed in panel IC of a top IC design house. Also co-authored the ICASSP 2024 paper (see Publications above) on a parallel medical-imaging deployment

Awards & Honors

1st Place + Best Presentation, 2023 Inventec Hackathon Competition Oct. 2023, Taipei

- 300+ participants, 74 teams. Cross-disciplinary partnership with an electromagnetics PhD on AI-powered laptop-antenna sensing/beamforming with zero hardware modifications; led the AI side

Talks & Presentations

Poster: "Do VLMs Need Vision Transformers?" — SUNY AI Symposium 2026 (SUNY-wide) Apr. 2026
Poster: "Glyph-Aware Text Embedding for Chinese LMs" — CSE 538 poster session, SBU May 2026
Poster: "Adaptive Chess Tutoring" — CSE 537 poster session, SBU May 2025
Course presentation: FlashAttention-1 paper, Topics in CS: VLM seminar, SBU Fall 2025
Guest Lecturer: NTUST Embedded DNN Processing course (Prof. Sheng-Chang Juan) Nov.–Dec. 2023

Teaching Experience

Teaching Assistant — CSE 582: Data Structures and Algorithms, Stony Brook University Fall 2024
Teaching Assistant — CSE 320: System Fundamentals II, Stony Brook University Spring 2025
Guest Lecturer, National Taiwan University of Science and Technology — see Talks above Nov.–Dec. 2023

Professional Service

Journal Reviewer: *Computers & Graphics* (2023); *ACM TOMM* (2023)

Technical Skills

Languages & ML: Python, C++, Verilog/SystemVerilog; PyTorch, Hugging Face (Transformers, PEFT, Datasets), State Space Models (Mamba family), Parameter-efficient fine-tuning (LoRA / dual-branch LoRA), Knowledge distillation, MLM pretraining, Flash Attention, bf16 mixed-precision, H100/H200 cluster training (SLURM)
Hardware & systems: RTL Design, AMD VCK5000 (AI Engine bring-up), Vitis/Vivado, RISC-V, LLVM IR, Linux/bash, Git, Docker. **Spoken**: Mandarin Chinese (native), English (professional). **Writing**: LaTeX

References available upon request.